

# Vasanth Bhaskara

Data Science at ZS | NIT Calicut  
Data Science and Applied AI for Real World Applications

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## EDUCATION

**B.Tech. in Mechanical Engineering**, Department **Rank 3** Dec 2020 – May 2024  
**National Institute of Technology (NIT)**, Calicut

- Cumulative GPA: **9.25/10.0**
- Relevant Coursework: Machine Learning, Statistics, Probability, Computer Programming

## FULL-TIME WORK EXPERIENCE

**Decision Analytics Associate** **ZS Associates, Pune**  
Data Science and Advanced Analytics for Pharmaceutical Applications Feb 2025 – Present

- Conducted marketing mix modeling for J&J's immunology brands enabling **data-driven omnichannel strategy optimization**
- Built **ETL pipelines** from ingestion of raw promotional data from various tables using **AWS redshift**, to various cleaning and aggregations using **Pandas** following the medallion architecture
- Developed **Linear** and **Bayesian regression** models with engineered features (e.g., ad stocking, normalization, log transforms) to accurately account for the impact of each marketing channel on total impactable sales
- Translated model outputs into deck-ready **insights** and excelled in **client communication**

**Condition Monitoring Engineer** **Hindustan Petroleum (HP), Visakhapatnam**  
Engineering for Large-Scale Critical Equipment Control and Monitoring Aug 2024 – Jan 2025

- Conducted equipment monitoring using software like Honeywell Forge and Bentley Nevada System

## TECHNICAL SKILLS

- Tech Stack:** Python, SQL, Databricks, Algorithms, Excel
- Frameworks:** PyTorch, XGBoost, PySpark, Pandas, NumPy

## SELECTED PROJECTS

**LLM Style Transfer via Parameter-Efficient Fine-Tuning** **Personal Project, Feb 2026**

- Fine-tuned **LiquidAI LFM2-1.2B** using **LoRA-based parameter-efficient fine-tuning** on 2,048 style-transfer examples to control generation style while training <0.05% of model parameters.
- Evaluated outputs using LLM-as-a-Judge methodology using **Gemini-2.5-Flash**, and **negative Log likelihood** metrics.
- Built **PyTorch** + HuggingFace pipelines for training, fine-tuning, and evaluation.

**LSTM Based Real-time Music Generation Application** **Personal Project, Jan 2026**  
Link: [Live Demo](#), [GitHub](#)

- Trained a character-level **LSTM** on ~800 songs with **embedding**, **sequence batching**, and **multinomial sampling** with **temperature control** to learn **next-token probability** distributions for symbolic music generation.
- Deployed** a **Streamlit**-based interactive **inference** system using pretrained weights with configurable generation parameters, real-time audio synthesis, and live web deployment.

**Promotion Impact Estimation using Bayesian Regression** **Decision Analyst, ZS, Dec 2025**

- Built **Bayesian regression** models to quantify channel-level contribution across marketing channels for a rapidly growing J&J's immunology brand.
- Delivered **actionable insights** enabling **data-driven** budget allocation, maximizing **ROI** and optimizing spend efficiency in competitive immunology markets.

## OTHER RELEVANT COURSES

- Verified Certifications:** Databricks Certified - Data Engineer Associate, [Link](#)
- Online Courses:** Stanford Online - Deep Learning Certificate, [Link](#)

## PUBLICATIONS

**Vasanth B S**, Vysakh P, M Thashreef, A P Sudheer. "Design, Fabrication and Control of a Multi-purpose Marine Robot", NIT Calicut, [Paper](#) (To be Submitted) May 2024